December 1970, by the launch of *Uhuru*, the first X-ray satellite (Figure 8.6c). Built by Giacconi's team, Uhuru contained a gas-filled, X-ray counting chamber one hundred times larger than the one they flew on their 1962 rocket. In front of the chamber's window were slats, like venetian blinds, to prevent the chamber from seeing X-rays from any direction except a few degrees around the perpendicular (Figure 8.6d). Uhuru, which discovered and cataloged 339 X-ray stars, was followed by several other similar but special-purpose X-ray satellites, built by American, British, and Dutch scientists. Then in 1978 Giacconi's team flew a grand successor to Uhuru: *Einstein*, the world's first true X-ray *telescope.* Because X-rays penetrate right through any object that they strike perpendicularly, even a mirror, the Einstein telescope used a set of nested mirrors along which the X-rays slide, like a tobogan sliding down an icy slope (Figures 8.6e,f). These mirrors focused the X-rays to make images of the X-ray sky 1 arc second in size-images as accurate as those made by the world's best optical telescopes (Figures 8.6g,h).

From Giacconi's rocket to the Einstein telescope in just sixteen years (1962 to 1978), a 300,000-fold improvement of angular accuracy had been achieved, and in the process our understanding of the Universe had been revolutionized: The X-rays had revealed neutron stars, black-hole candidates, hot diffuse gas that bathes galaxies when they reside in huge clusters, hot gas in the remnants of supernovae and in the coronas (outer atmospheres) of some types of stars, and particles with ultra-high energies in the nuclei of galaxies and quasars.

Of the several black-hole candidates discovered by X-ray detectors and X-ray telescopes, Cygnus X-1 (Cyg X-1 for short) was one of the most believable. In 1974, soon after it became a good candidate, Stephen Hawking and I made a bet; he wagered that it is not a black hole, I that it is.

Carolee Winstein, whom I married a decade after the bet was made, was mortified by the stakes *(Penthouse* magazine for me if I win; *Private Eye* magazine for Stephen if he wins). So were my siblings and mother. But they didn't need to worry that I would actually win the *Penthouse* subscription (or so I thought in the 1980s); our information about the nature of Cyg X-1 was improving only very slowly. By 1990, in my view, we could be only 95 percent confident it was a black hole, still not confident enough for Stephen to concede. Evidently Stephen read the evidence differently. Late one night